

# Patrick FERNANDES

## PERSONAL DATA

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## ACADEMIC QUALIFICATIONS

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JUNE 2018 Master of Philosophy in ADVANCED COMPUTER SCIENCE  
**University of Cambridge**, Cambridge  
Distinction

JULY 2017 Bachelor of Science in COMPUTER ENGINEERING  
**Instituto Superior Técnico**, Lisbon  
Average Grade: 19/20

JULY 2014 High School Degree in SCIENCE  
**Escola Secundária "Ferreira Dias"**, Sintra  
Average Grade: 18/20

## PUBLICATIONS

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Under Review | STRUCTURED NEURAL SUMMARIZATION  
**P. Fernandes**, M. Allamanis, M. Brockschmidt.  
[arXiv pre-print 2018](#)

## WORK EXPERIENCE

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*Current* | AI Resident at MICROSOFT RESEARCH, Cambridge  
*Deep Program Understanding*  
Currently work as an AI Resident, a new 1-year training program designed to train and prepare the residents to be Researchers or Research Engineers in the AI area. Will be working on two different 6-month projects throughout my residency.

SUMMER 2018 | Research Internship at MICROSOFT RESEARCH, Cambridge  
*Deep Program Understanding*  
Worked in trying to learn better distributed representations for source code functions using graph neural networks by exploiting control and data flow information extractable from the parse trees. Also developed a novel graph neural network model that is able to exploit a sequencing in the graph and applied this model to source code and natural language summarization

OCTOBER 2015  
- JUNE 2016 | Researcher at CODACY, Lisbon  
*Machine Learning for Static Code Analysis*  
Due to a good performance in the intership at INESC-ID, was invited to do some research for CODACY, a company that focus on static analysis tools for companies and developers. Our research group focus on possible applications of machine learning algorithms for automatic code analysis task where "standard" algorithms struggle achieve good results.

SUMMER 2015 | Summer Intern at INESC-ID, Lisbon  
*Natural Language Processing*  
Applied for a summer intership in my first year, and did some research on the application of DNA alignment algorithms to automatic audio extraction from movies, for a easier creation of labelled audio datasets to use in NLP projects

## AWARDS

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- SEPT. 2016    Excellency Award for outstanding performance on the academic year 2015/2016.  
AUGU. 2016    Excellency Award for outstanding performance on the academic year 2014/2015.  
SEPT. 2014    Ferreira Dias' Best Student Award of the year 2013/2014.

## ACTIVITIES

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### SECTT:

- IST's cybersecurity team
- Participed regularly in CTF events
- Rated #1 security team in Portugal at the tme.

### SCIKIT-LEARN:

- Open-source machine learning project
- Contribute by fixing bugs and adding new features

## SKILLS AND INTERESTS

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LANGUAGES: Portuguese (Native); English(Fluent, IELTS Band 8); Spanish (Intermediate);

TECHNICAL SKILLS: Python (tensorflow, pytorch, scikit, numpy) , Matlab, Java, C++, C

RESEARCH INTERESTS: Machine Learning, Representation Learning, Program Understanding, Reinforcement Learning, Theorem Proving